



FumiGuard Pro

Pest Control & Fumigation Management System
Web Platform + Mobile Worker App

Submitted by	Abdullah Haroon
Live Web App	https://fumication-app-web.vercel.app
Live API	https://fumication-app.vercel.app
Repository	github.com (Fumication-app)
Date	June 2026

1. Project Title & Description

FumiGuard Pro is a full-stack digital platform that replaces paper-based workflows for the pest-control / fumigation industry. It lets a fumigation company manage clients, schedule jobs, dispatch field workers, track GPS-verified site check-ins, manage chemical inventory, and generate signed service reports — all from a single system.

The project consists of three parts that share one PostgreSQL database and one authentication system:

- **Web App (Next.js)** — the admin/manager/client dashboard, used to schedule jobs, manage workers, locations and inventory, and review/generate PDF service reports.
- **API (Express + PostgreSQL)** — a REST API providing authentication, role based access control, and CRUD endpoints for all entities.
- **Mobile App (React Native / Expo)** — "FumiGuard Worker" — a native Android/iOS app for field workers to view assigned jobs, check in via GPS at the job site, and fill out the service report on-site.

2. Objectives & Scope

Objectives

- Digitize the entire fumigation service lifecycle — from job scheduling to signed service report — removing paper forms.
- Provide role-based dashboards so admins, managers, field workers and clients each see only what is relevant to them.
- Verify that a worker actually visited the job site using GPS-based geofenced check-ins.
- Track chemical/equipment inventory and warn when stock falls below a threshold.
- Generate a standardized, downloadable PDF "Service Report" matching the industry's IPM (Integrated Pest Management) documentation format.
- Give field workers a lightweight mobile app to do everything they need on-site without a laptop.

Scope

- In scope: authentication & RBAC, job scheduling & assignment, GPS check-ins, locations (client premises) with map picker, inventory & warehouses, structured service-report form + PDF export, notifications, PWA support for the web app, and a companion mobile app for workers.
- Out of scope (future work): payments/invoicing UI (the `invoices` table exists in the schema but has no dedicated route/UI yet), SMS notifications, multi-tenant company support.

3. Implementation Details

Architecture

The project is a **Turborepo monorepo** named `fumiguard-pro` using npm workspaces (`apps/*` , `packages/*`), with the standalone Expo mobile app kept outside the workspace for easy isolation.

```
Fumication-app/
├─ apps/
│  └─ web/      → Next.js dashboard (apps/web)
│     └─ api/    → Express REST API (apps/api)
├─ packages/    → shared packages
├─ mobile/      → Expo React Native app "FumiGuard Worker" (standalone)
└─ turbo.json   → Turborepo task pipeline
```

Tech Stack

Layer	Technology
Web Frontend	Next.js 16, React 19, Tailwind CSS v4, TanStack Query 5, Axios, Leaflet (maps), jsPDF + jspdf-autotable (PDF export), lucide-react icons, TypeScript
Backend API	Express 4, PostgreSQL (pg driver), JWT (jsonwebtoken), bcryptjs, Zod validation, Helmet, express-rate-limit, CORS, Firebase Admin SDK
Database	Neon (managed cloud PostgreSQL)
Mobile App	Expo SDK 54, React Native 0.81, React 19, React Navigation 7, AsyncStorage, expo-location
Hosting	Vercel — web app and API both deployed as Vercel projects (API as a serverless function)

Authentication & Role-Based Access Control (RBAC)

Authentication uses JWT tokens issued by `/api/auth/login` and verified by middleware on every protected route. Four roles are supported:

Role	Access
admin	Full access to every module: dashboard, jobs, workers, locations, inventory, reports, notifications.
manager	Same as admin — full operational access for day-to-day management.
worker	Restricted to Dashboard, Jobs (their own assignments) and Reports — matches the mobile app's feature set.
client	Restricted to Reports — clients can only view/download their service reports.

Database / API Entities

The API exposes REST routes for the following entities (8 route modules):

- `auth` — register/login, returns JWT + user profile
- `users` — worker/user management
- `locations` — client premises (with GPS coordinates)
- `jobs` + `job_assignments` — scheduling and assigning workers to jobs, status transitions (scheduled → in-progress → completed)
- `checkins` — GPS check-in/out events tied to a job + worker
- `inventory` + `warehouses` — chemical stock levels, thresholds and suppliers
- `reports` + `report_chemicals` — the structured service report (JSONB `form_data`) plus signatures
- `notifications` — in-app notification feed

Deployment

Both the web app and the API are deployed on **Vercel**:

- **API** (`https://fumication-app.vercel.app`) — the Express app is exported and wrapped by a thin `apps/api/api/index.ts` serverless handler; `vercel.json` rewrites every request to it.
- **Web** (`https://fumication-app-web.vercel.app`) — standard Next.js deployment with `NEXT_PUBLIC_API_URL` pointing at the API above.
- **Mobile** — built as a standalone Android APK via EAS Build (`eas build --platform android --profile preview`), pointed at the same production API.

4. Features & Functionalities

Dashboard

At-a-glance overview: total/completed jobs, low-stock inventory alerts, worker count, pending jobs, and a "Recent Jobs" activity table.

Job Scheduling

Create jobs against a location, assign one or more workers, track status (scheduled → in-progress → completed) and view time taken.

GPS Check-Ins

Workers check in/out from the mobile app; the system records their GPS coordinates and shows an "On site" badge with a map link on the job detail page.

Locations

Client premises with full address and GPS pin, set via an interactive Leaflet map picker with address search (Geoapify).

Inventory Management

Chemicals/equipment tracked by quantity, warehouse, threshold and supplier, with automatic OK / Low-stock status and dashboard alerts.

Service Reports (PDF)

A structured IPM-format report — pesticides used, application method, target pests, area remarks, and a full monitoring trap grid (glue/live/mouse traps) — downloadable as a formatted PDF (via jsPDF) with worker & client signatures.

Worker Management

Admins/managers can add workers, view their status (active), contact info and join date.

Notifications

In-app notification bell with unread indicator for job/system events.

Progressive Web App

The web dashboard ships a manifest + service worker, so it can be installed and used offline-tolerantly like a native app.

Mobile Worker App

"FumiGuard Worker" (Expo/React Native): login, view assigned jobs, GPS check-in/out at the job site, and fill the same structured service report form on-site.

5. Instructions for Running the Project

Option A — Use the Live Deployment (fastest)

Component	URL	Notes
Web Dashboard	https://fumication-app-web.vercel.app	Login with seeded admin account
API	https://fumication-app.vercel.app	<code>/health</code> returns service status
Demo Login	<code>admin@fumiguard.com / admin123</code>	Role: admin (full access)

Option B — Run Locally

Prerequisites: Node.js 18+, npm, a PostgreSQL database (the project uses a Neon cloud Postgres instance — connection string supplied via env var).

```
# 1. Install dependencies (root of monorepo)
npm install

# 2. Configure environment variables
#   apps/api/.env
#     DATABASE_URL=postgres://...
#     JWT_SECRET=your-secret
#   apps/web/.env.local
#     NEXT_PUBLIC_API_URL=http://localhost:5000
#     NEXT_PUBLIC_GEOAPIFY_KEY=your-geoapify-key

# 3. Run both API and Web in dev mode (via Turborepo)
npm run dev
#   API    → http://localhost:5000
#   Web    → http://localhost:3000
```

Option C — Mobile App (FumiGuard Worker)

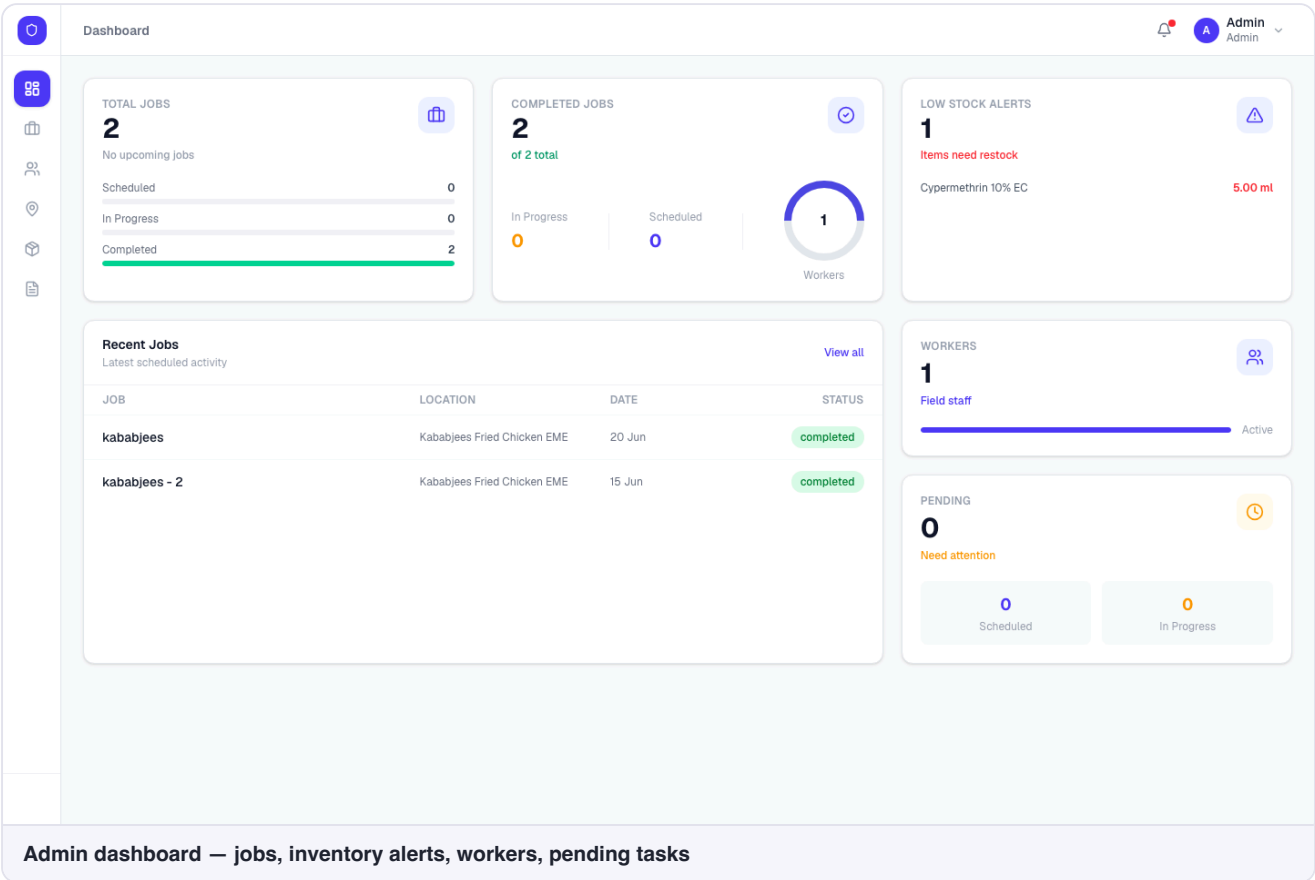
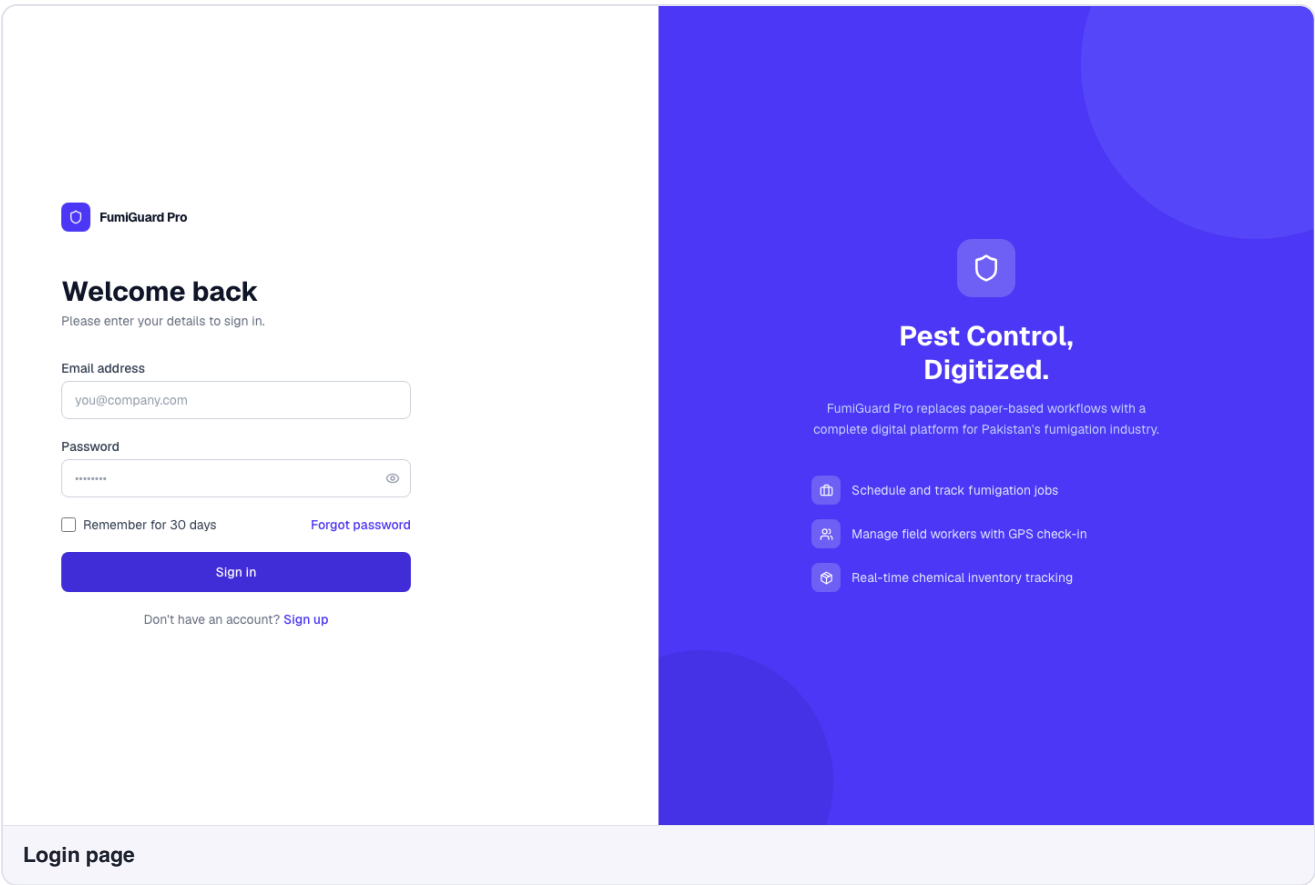
```
cd mobile
npm install

# Point the app at an API (mobile/src/lib/config.ts)
#   - Production: https://fumication-app.vercel.app (default, works anywhere)
#   - Local dev:  http://<your-LAN-IP>:5000

# Run in Expo Go for development
npx expo start

# Or build a standalone Android APK
npx eas-cli login
npx eas-cli build --platform android --profile preview
```

6. Screenshots — Web Application



6. Screenshots — Web Application (continued)

Jobs

Admin

Admin

Jobs

Schedule and manage fumigation tasks

+ New Job

Search jobs or locations...

All statuses

2 results

JOB	LOCATION	SCHEDULED	STATUS	
kababjees	Kababjees Fried Chicken EME	20 Jun 2026	completed	View
kababjees - 2	Kababjees Fried Chicken EME	15 Jun 2026	completed	View

Jobs list — schedule and track fumigation tasks

Jobs

Admin

Admin

<

kababjees

completed

Kababjees Fried Chicken EME

Delete

Job Details

LOCATION

Kababjees Fried Chicken EME

SCHEDULED

Sat, 20 June 2026 at 02:47 am

STARTED

12 Jun 2026, 07:45 pm

COMPLETED

13 Jun 2026, 03:19 pm

TIME TAKEN

19 hr 33 min

Worker Check-ins

GPS locations sent by workers on-site

A

Abdullah Haroon

bsai24003@itu.edu.pk

On site

12 Jun, 07:46 pm

31.43842500, 74.21101400 — View on map

Assigned Workers

1 assigned

A

Abdullah Haroon

bsai24003@itu.edu.pk

Report

[View / download report](#)

Job detail — GPS check-ins, assigned workers, report link

6. Screenshots — Web Application (continued)

Jobs

Admin

Admin

Service Report

kababjees — Kababjees Fried Chicken EME

Visit Details

DATE

13/06/2026

CLIENT

Kababjees Fried Chicken EME

TIME IN

07:45 PM

TIME OUT

02:58 PM

SERVICE ADDRESS

Plot # 1, Block A, Rajpoot Cooperative Housing Society, Canal Road, near DHA-EME, Lahore, Pakistan

Routine

Follow Up

Complaint

Section I. Pesticide Application

As per IPM basic principle, use pesticides only when no other options are left.

PESTICIDE / APP. RATE

☒ Alphaguard 10% SC

200/20

☐ Topbait gold gel

spots/m3

☐ Fly Bait

gm/SFT

METHOD

☐ Injecting (I)

☒ General Spraying (IRS)

☐ Spot Treatment (SP)

☐ ULV Fogging (ULV)

☐ Thermal Fogging (TF)

☒ Dusting (D)

☐ Gel App./Insect Baiting (B)

☐ Aviciding (AV)

☐ Soil Injecting (SI)

TARGET

☒ Mosquitoes

☒ Flies

☒ Roaches

☐ Birds

☐ Termites

☐ Bed Bugs

☐ Wasps

☐ Moths

☐ Bees

Service report form — pesticides, methods, targets and monitoring

Workers

Admin

Admin

Workers

Field staff and their assignment status

+ Add Worker

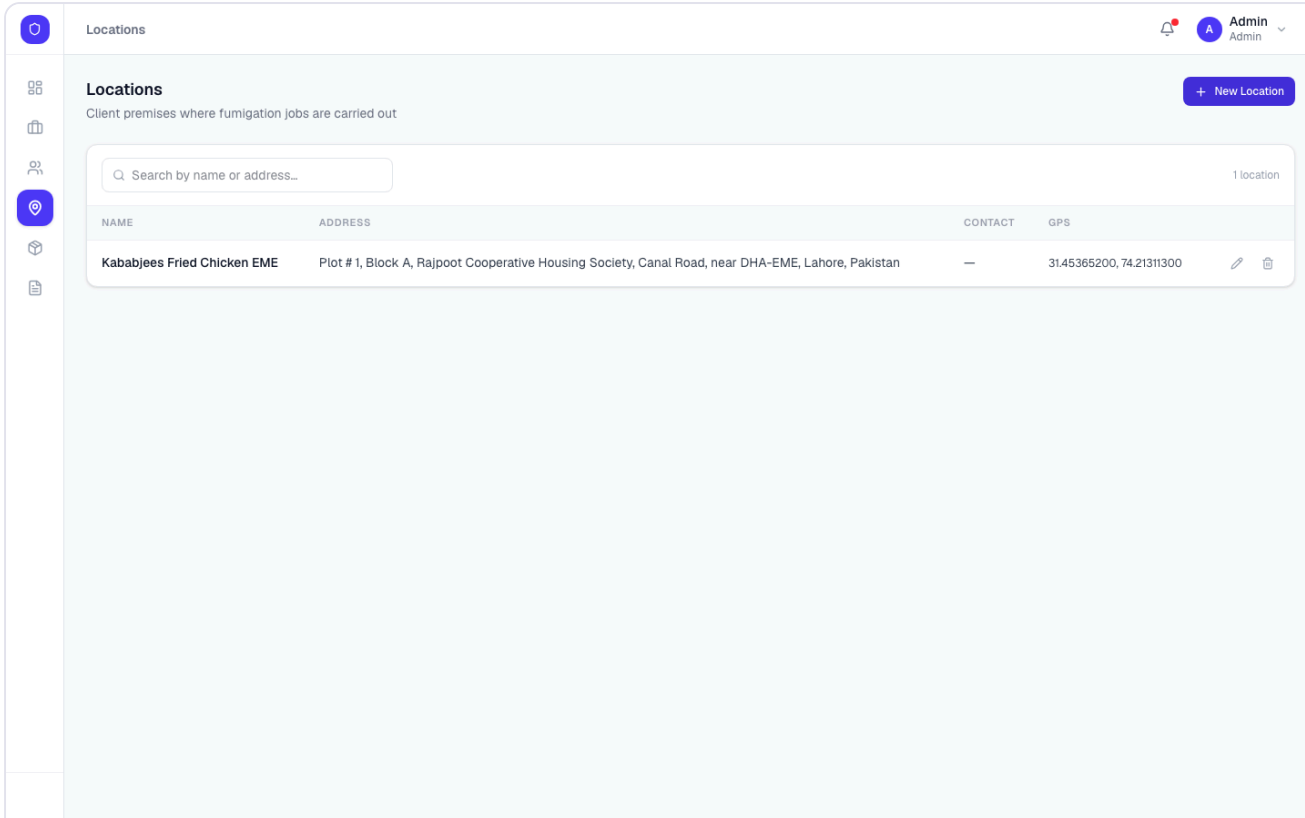
Q Search by name or email...

1 worker

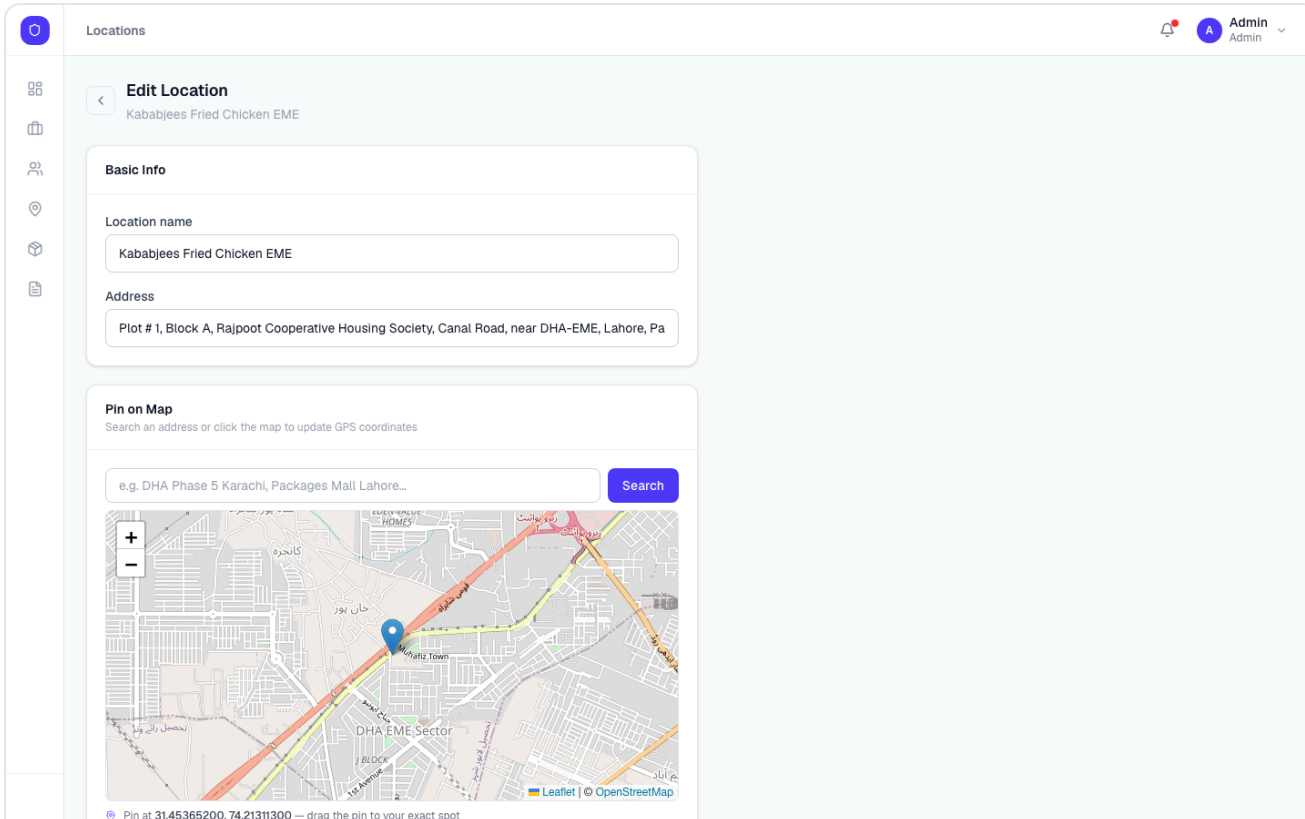
NAME	EMAIL	PHONE	STATUS	JOINED
A Abdullah Haroon	bsai24003@itu.edu.pk	03262446260	Active	23 May 2026

Workers — field staff and assignment status

6. Screenshots — Web Application (continued)



Locations — client premises list



Edit location — map picker with GPS pin

6. Screenshots — Web Application (continued)

Inventory

Admin Admin

Inventory

Chemicals and equipment stock levels

Search Items or warehouse...

1 Item

ITEM	WAREHOUSE	QUANTITY	THRESHOLD	SUPPLIER	STOCK	
Cypermethrin 10% EC Updated desc	—	5.00 ml	10.00 ml	AgriChem	OK	

Inventory — chemical stock levels and thresholds

Reports

Admin Admin

Reports

Fumigation job reports and digital records

JOB	LOCATION	WORKER	PESTS FOUND	DATE	SIGNATURES		
kababjees - 2	Kababjees Fried Chicken EME	Abdullah Haroon	Mosquitoes, SGP	14 Jun 2026	Worker	Client	View / Download
kababjees	Kababjees Fried Chicken EME	Abdullah Haroon	Mosquitoes, Flies, Roaches, SGP	13 Jun 2026	Worker	Client	View / Download

Reports list — all service reports with signature status

6. Screenshots — Web Application (continued)

○

Reports

Admin

Admin

Service Report

Print

Download PDF

SERVICE REPORT

Document # IFPCS-IPM-13
Revision # 01
Revision Date 01-January-24
Effective Date From 02-January-24

Date:13 June 2026

Time In:19:45

Time Out:14:58

Client:Kababjees Fried Chicken EME

Service Address:Plot # 1, Block A, Rajpoot Cooperative Housing Society, Canal Road, near DHA-EME, Lahore, Pakistan

ROUTINE

FOLLOW UP

COMPLAINT

Section I. Pesticide Application

As per IPM basic principle, use pesticides only when no other options are left. Pesticides are supplement to IPM plan not stand alone solutions. Use pesticides to rescue emergencies and work on root causes. Ensure pesticide safety by pre & post notifications.

PESTICIDE

APP. RATE

METHOD

TARGET

AREA/REMARKS

☒ Alphaguard 10% SC

☐ Topbait gold gel

☐ Fly Bait

☐ Permethrin 0.5% powder

200/20

—

—

☐ Injecting (I)

☒ General Spraying (IRS)

☐ Spot Treatment (SP)

☐ ULV Fogging (ULV)

☐ Thermal Fogging (TF)

☒ Dusting (D)

☐ Gel App./Insect Baiting (B)

☐ Aviciding (AV)

☐ Soil Injecting (SI)

☒ Fumigation (F)

☒ rodent poison

☐ Mosquitoes

☒ Flies

☒ Roaches

☐ Birds

☐ Termites

☐ Bed Bugs

☐ Wasps

☐ Moths

☐ Bees

☒ SGP

in side

Time of Application:

Follow up (If Req.):

Reason if not done:

Section II. Monitoring & Servicing

Status: ☒ Okay ☐ Damage ☐ Lost ☐ N/A

Action: ☒ No Need ☐ Replaced ☐ New ☐ Unmount ☐ Mounted

Glue Trap

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

Status

Okay

Okay

Okay

Okay

Okay

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

Count

0

0

0

0

0

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

Action

No Need

No Need

No Need

No Need

No Need

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

Live Trap

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

Status

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

Count

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

Action

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

—

Mouse Trap

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

Generated Service Report — downloadable as PDF

7. Additional Information

Notable Engineering Details

- API returns PostgreSQL `numeric` columns as strings; the mobile app explicitly converts these (`Number(value)`) before formatting GPS coordinates.
- The Express API is reused as-is for both local development (`app.listen`) and Vercel serverless deployment (`export default app`), guarded by `require.main === module` .
- The mobile app shares its service-report data model (`reportForm.ts`) with the web app, so a report started on mobile can be completed/edited on web and vice versa.

Mobile App — FumiGuard Worker

A companion Expo/React Native app gives field workers a focused subset of the system on their phones:

- **Login** — same JWT-based auth as the web app.
- **Jobs** — list of jobs assigned to the logged-in worker, with status.
- **Job Detail** — GPS check-in/check-out at the job location (geofenced), live status transitions.
- **Service Report** — fill out or edit the same structured IPM report used on the web, directly from the job site.

Repository Structure Recap

Path	Purpose
<code>apps/web</code>	Next.js dashboard (all roles)
<code>apps/api</code>	Express REST API + PostgreSQL access layer
<code>mobile</code>	Expo "FumiGuard Worker" Android/iOS app
<code>render.yaml</code>	Legacy Render blueprint (superseded by Vercel deployment)

Summary

FumiGuard Pro demonstrates a complete, deployed, full-stack product: a relational data model shared across a web dashboard, a REST API, and a native mobile app, with real-world features such as role-based access control, GPS-verified field operations, inventory alerting, and regulator-style PDF report generation — all running live on Vercel against a managed PostgreSQL database.